

SE Lab - 1

Household Carbon Footprint & Target Tracker

SRN:PES1UG25CS836

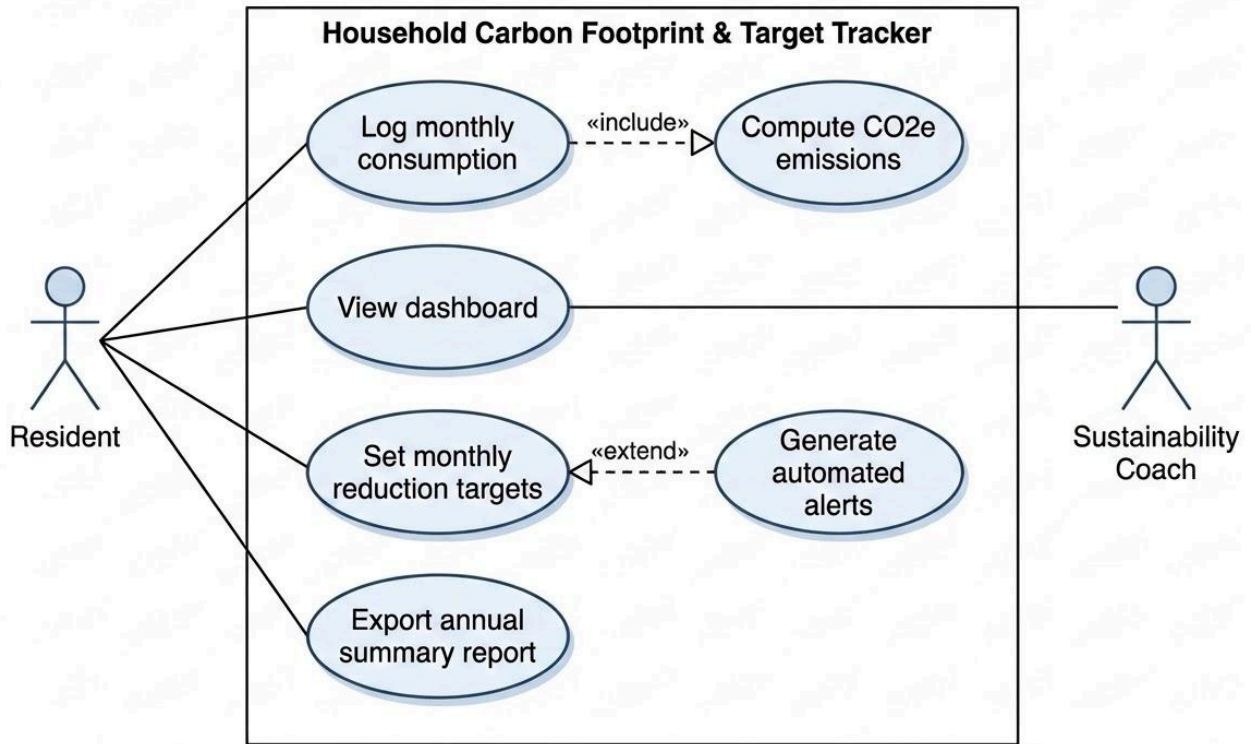
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Requirements Table:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO2e) emissions.	High	Pass: Calculated emissions match EPA conversion factor formulas; Fail: Zero or negative emission values generated.	Essential for baseline tracking.
FR-002	Functional	The system shall provide a user dashboard to view monthly carbon footprints compared against regional averages.	Medium	Pass: Dashboard loads within 1s and displays correct comparative data; Fail: Data mismatch or latency > 2s.	Encourages user engagement through benchmarking.
FR-003	Functional	The system shall allow users to set monthly reduction targets for CO2e emissions.	High	Pass: Target is saved to database and reflected in user profile; Fail: Save failure or invalid input validation.	Direct interaction for goal setting.

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FR-004	Functional	The system shall generate automated alerts when consumption exceeds the set target by 10%.	Medium	Pass: Notification triggered immediately upon submission; Fail: Alert failure or incorrect calculation threshold.	Proactive user guidance.
FR-005	Functional	The system shall export an annual summary report in PDF format for user records.	Low	Pass: Valid PDF generated with accurate data; Fail: File corruption or data truncation.	Necessary for long-term record keeping.
NFR-001	Performance & Security	The annual carbon reduction milestone chart must render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures high responsiveness for data visualization.
NFR-002	Reliability	The system shall maintain 99.9% uptime for data entry and reporting services.	High	Pass: Server logs confirm < 9 hours downtime per year; Fail: Extended service outages.	Vital for consistent user logging habits.

Use-Case Diagram:



Use-Case Flow Specification:

Use Case: Log Monthly Consumption

Actors: Resident

Preconditions:

- The resident has an active account.
- The resident has logged in to the dashboard.
- Consumption data (electricity, transport) for the period is available.

Postconditions:

- The system has recorded the usage logs.
- The system has computed and updated the monthly CO2e emission value.
- The updated carbon footprint is reflected on the user's dashboard.

Main Success Scenario:

1. Resident navigates to the 'Log Consumption' section of the dashboard.
2. Resident enters monthly electricity consumption (kWh).
3. Resident enters monthly transport distance (km).
4. Resident submits the data.
5. System validates the input data.
6. System calculates the total monthly carbon equivalent (CO2e) emissions (includes 'Compute CO2e emissions' use case).
7. System saves the consumption logs and calculated emissions to the database.
8. System updates the user's dashboard with the new monthly emission figures.

Alternate Flow:

1. 1a. Invalid or negative input detected by the system:
 - The system displays an error message prompting the user to re-enter valid data.

The system clears the invalid fields and waits for user correction.